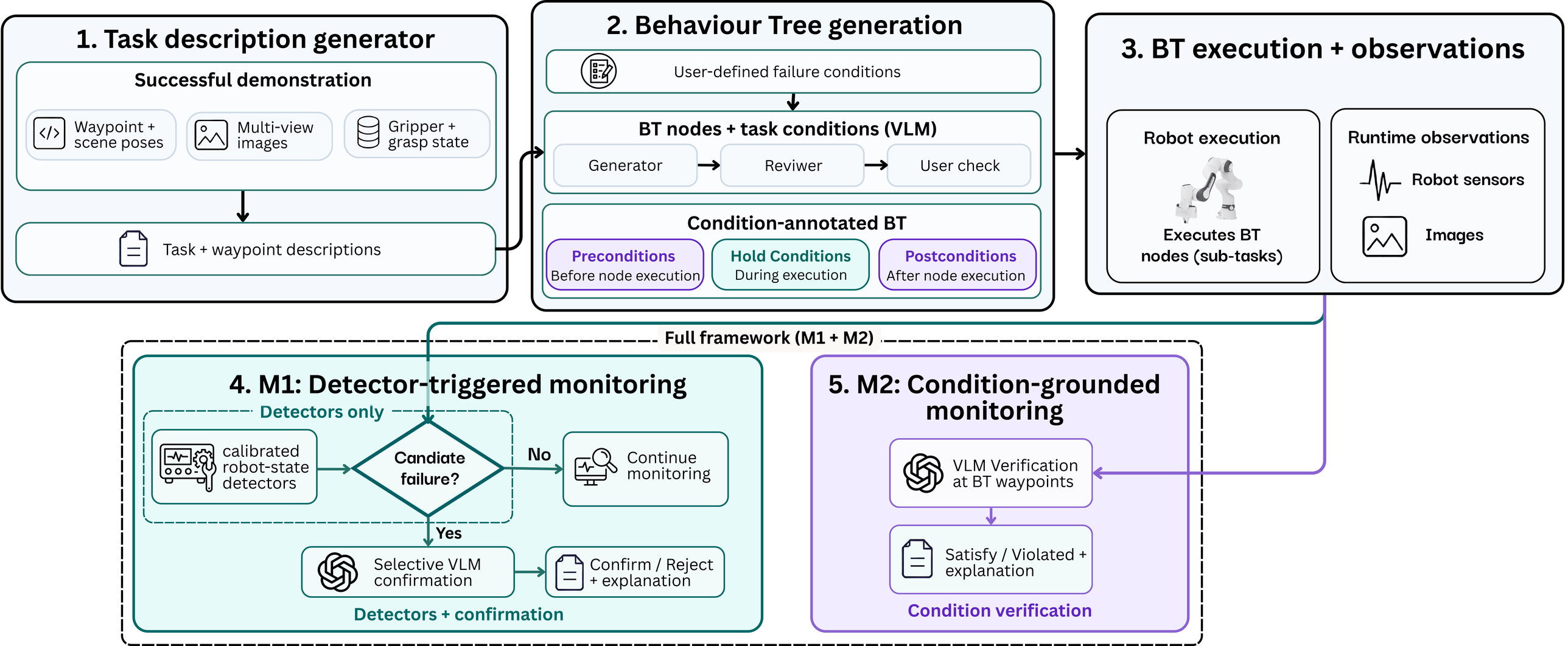

1. Task description generator
Successful demonstration
Waypoint + scene poses
Multi-view images
Gripper + grasp state
Task + waypoint descriptions
2. Behaviour Tree generation
User-defined failure conditions
BT nodes + task conditions (VLM)
Generator
Reviwer
User check
Condition-annotated BT
Preconditions
Before node execution
Hold Conditions
During execution
Postconditions
After node execution
3. BT execution + observations
Robot execution
Executes BT nodes (sub-tasks)
Runtime observations
Robot sensors
Images
Full framework (M1 + M2)
4. M1: Detector-triggered monitoring
Detectors only
calibrated robot-state detectors
Candiate failure?
No
Continue monitoring
Yes
Selective VLM confirmation
Confirm / Reject + explanation
Detectors + confirmation
5. M2: Condition-grounded monitoring
VLM Verification at BT waypoints
Satisfy / Violated + explanation
Condition verification